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 Oefeningenreeks 1D nr. 23.5

$$A(z) = 2z^3 - (5 + i)z^2 + (3 + 3i)z - (2 + 2i)$$

$$\begin{array}{c|ccc} 2 & 2 & -5-i & 3+3i & -2-2i \\ & & 4 & -2-2i & 2+2i \\ \hline & 2 & -1-i & 1+i & 0 \end{array}$$

$$\Rightarrow A(z) = (z - 2)(2z^2 - (1 + i)z + 1 + i)$$

Vind nulpunten van $(2z^2 - (1 + i)z + 1 + i)$

$$\begin{cases} x^2 - y^2 = -8 \\ 2xy = -6 \end{cases}$$

$$\begin{cases} x^2 - y^2 = -8 \\ x^2(-y^2) = -9 \end{cases}$$

De resolvente vierkantsvergelijking:

$$\lambda^2 - 8\lambda - 9 = 0$$

$$D = 64 - 4 \cdot (-9)$$

$$D = 100$$

$$\lambda_{1,2} = \frac{8 \pm \sqrt{100}}{2}$$

$$x = \pm 1 \text{ en } y = \pm 3$$

$$b < 0:$$

$$D = (1 - 3i) \text{ of } D = (-1 + 3i)$$

Oplossingen voor z :

$$z_{2,3} = \frac{1 + i \pm (1 - 3i)}{4}$$

$$z_2 = i$$

$$z_3 = \frac{1 - i}{2}$$

$$\Rightarrow A(z) = (z - 2)(z + i)(2z - 1 - i)$$

References